Quantification: 548_hyp_HP_z14c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	14
frame-edge lacuna count	7

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2580.29	1753	1930.27	px^2
major_axis_length_px	lacuna shape	78.7726	72.0264	32.9493	px
minor_axis_length_px	lacuna shape	43.3227	38.318	18.3368	px
aspect_ratio	lacuna shape	1.8566	1.9787	0.3968	unitless
eccentricity	lacuna shape	0.8092	0.8629	0.107	unitless
solidity	lacuna shape	0.7901	0.8075	0.0785	unitless
roots_count	headline	9	7	5.1141	count
ring_length_r30_px	headline	291.429	257.5	149.968	px
ring_length_r60_px	reported	600.929	568	265.205	px
owned_length_px	ownership-dependent	1020.24	848.995	775.312	px
edge_count	ownership-dependent	51.5714	35.5	39.2775	count
mean_edge_length_px	ownership-dependent	20.6407	19.6628	4.9564	px
perimeter_px	normalised	268.524	221.622	137.582	px
ring_area_r30_px2	normalised	8680.93	7623	3718.35	px^2
ring_area_r60_px2	normalised	19135.9	18514.5	7625.53	px^2
in_frame_fraction_r30	normalised	0.9773	1	0.0559	unitless
in_frame_fraction_r60	normalised	0.9533	1	0.0995	unitless
ring_density_r30	normalised	0.0329539	0.0319377	0.00565347	px^-1
ring_density_r60	normalised	0.0313662	0.031802	0.00479275	px^-1
roots_per_100px_perimeter	normalised	3.537	3.25	1.4689	per 100 px
ring_attached_length_r30_px	network v2	262	221.5	150.649	px
ring_attached_length_r60_px	network v2	468.5	410	252.245	px
ring_length_w_r30_px	network v2	333.982	297.179	171.264	px
ring_length_w_r60_px	network v2	685.891	643.701	303.876	px
sholl_crossings_r10	network v2	6.8571	6	3.1588	count
sholl_crossings_r20	network v2	8.2143	7.5	4.2999	count
sholl_crossings_r30	network v2	8	6.5	4.3677	count
loop_count	loops, report only	3.5714	2.5	3.5673	count
loop_edge_share	loops, report only	0.2103	0.2424	0.1219	share
ring_width_mean_r30_px	width	7.8066	7.6726	0.5399	px

Per field
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measure	value	units
analysed_area_px2	991765	px^2
total_skeleton_length_px	27567	px
canalicular_length_density_per_px	0.0277959	px^-1
junction_count	584	count
junction_density_per_px2	0.00058885	px^-2
lacuna_count	21	count

lacunae_per_px2	2.117e-05	px^-2
skeleton_component_count	382	count
mean_component_length_px	72.1649	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0316315	px^-1
field_density_without_flagged_per_px	0.0277959	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	248	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0277959	px^-1
field_length_density_with_unattached_per_px	0.028046	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	76	count
loop_edge_share	0.1975	share
edge_count	1519	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8715	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27567	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
4	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
5	996	48.8421	30.1974	1.6174	0.786	0.83	-0.3922	177.571	118.507	False
6	1527	72.7312	32.1897	2.2595	0.8967	0.7689	1.1031	215.125	119.538	False
7	791	45.4603	27.379	1.6604	0.7983	0.777	-1.1935	276.899	15.5398	True
8	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
9	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
10	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
11	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
12	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
13	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
14	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
15	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
16	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
17	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
18	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
19	3821	95.6808	52.75	1.8139	0.8343	0.9065	-1.1603	884.495	983.684	True
20	1979	71.3216	39.2045	1.8192	0.8354	0.8154	-1.307	916.707	906.786	False
21	8745	138.777	89.9937	1.5421	0.7612	0.8115	1.2047	976.542	708.753	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	191	343	781.281	31	25.2026	6.9981
2	10	354	744	880.637	44	20.0145	7.8088
3	10	241	509	1773.19	91	19.4856	7.7676
4	7	288	596	884.205	27	32.7483	8.7175
5	4	123	248	551.132	27	20.4123	8.0682
6	6	172	421	613.044	22	27.8656	7.5689
7	3	151	227	63.1838	2	31.5919	5.7979
8	6	130	271	74.7696	3	24.9232	7.3539
9	12	321	689	1428.11	80	17.8514	7.3687
10	9	469	961	2883.77	138	20.8969	8.7134
11	10	353	836	972.323	52	18.6985	7.1274
12	2	98	214	367.463	18	20.4146	7.3659
13	4	218	399	463.345	22	21.0611	7.4299
14	7	184	566	283.451	21	13.4977	7.9669
15	22	607	1128	1944.31	98	19.8399	7.5777
16	7	274	570	813.784	44	18.4951	7.2242
17	8	311	574	1424.43	70	20.3489	7.755
18	17	525	861	1278.74	78	16.3942	7.831
19	6	201	375	404.09	23	17.5691	8.0704
20	5	175	358	319.208	19	16.8004	8.5769
21	7	428	897	763.896	40	19.0974	6.8574

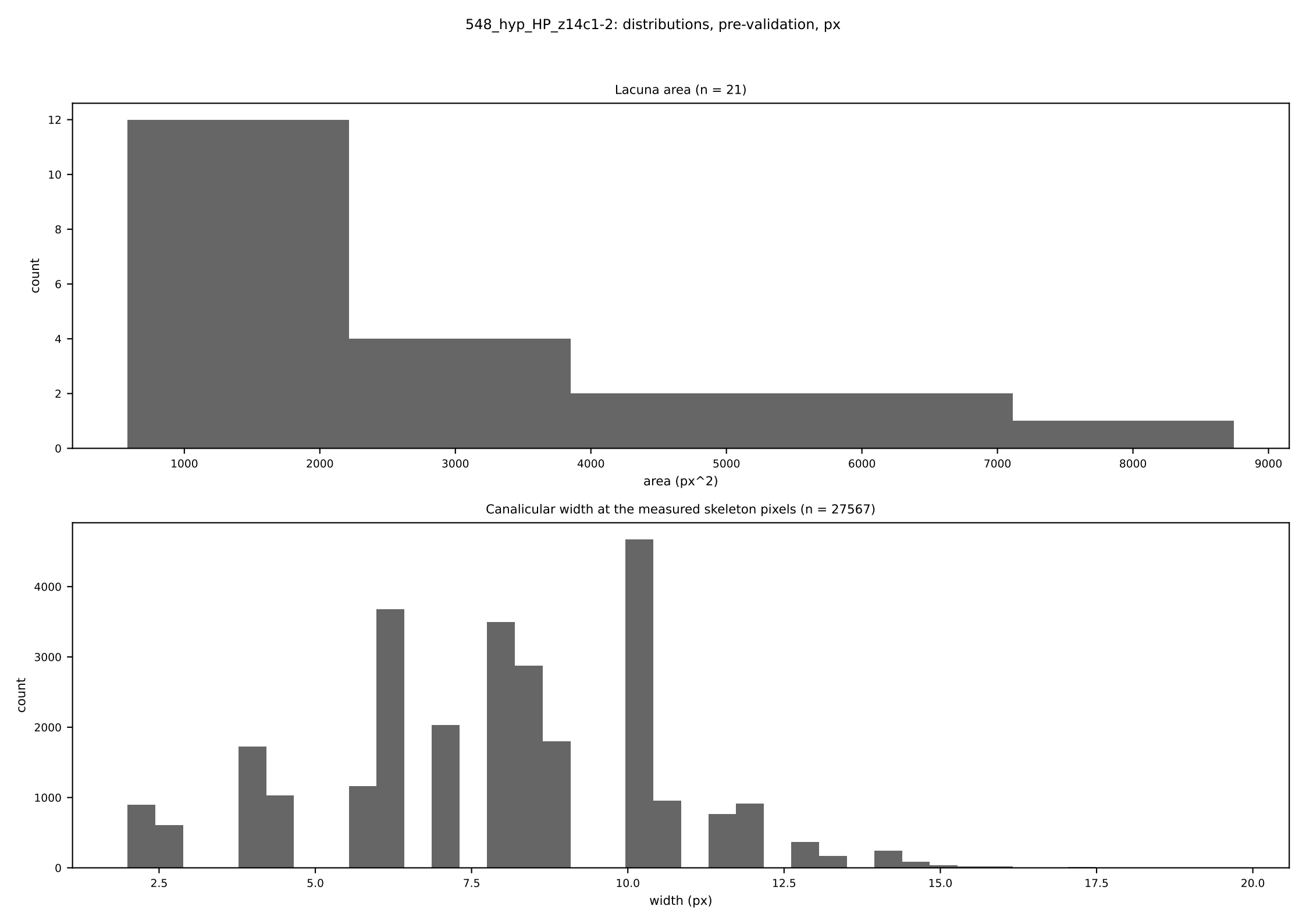
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6967	12578	0.7247	0.6734	0.027415	0.0272698	2.2231
2	330.693	8703	19577	0.7916	0.7301	0.0406756	0.0380038	3.0239
3	289.563	8243	15900	0.836	0.748	0.0292369	0.0320126	3.4535
4	351.161	12196	26656	0.9903	0.8855	0.0236143	0.0223589	1.9934
5	192.989	4006	9474	1	1	0.0307039	0.0261769	2.0727
6	221.38	6390	14742	1	1	0.0269171	0.0285579	2.7103
7	143.74	3169	5742	0.6661	0.6142	0.0476491	0.0395333	2.0871
8	148.61	4205	7189	0.8558	0.7127	0.0309156	0.0376965	4.0374
9	184.652	7467	20230	1	1	0.0429891	0.0340583	6.4987
10	352.919	11920	25165	1	1	0.0393456	0.038188	2.5502
11	328.233	11389	28087	1	1	0.0309948	0.0297647	3.0466
12	193.539	5102	10952	0.6366	0.5949	0.0192082	0.0195398	1.0334
13	137.196	6367	12630	1	1	0.0342391	0.0315915	2.9155
14	104.569	5596	16799	1	1	0.0328806	0.0336925	6.6942
15	631.595	16663	33826	1	1	0.036428	0.0333471	3.4832
16	199.865	7779	21047	1	1	0.035223	0.0270822	3.5024
17	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
18	394.747	12775	22894	1	1	0.0410959	0.0376081	4.3066
19	281.38	6112	10015	0.7455	0.6918	0.0328861	0.0374438	2.1323
20	221.865	6537	13263	1	1	0.0267707	0.0269924	2.2536
21	525.044	12017	25181	0.7555	0.698	0.0356162	0.0356221	1.3332

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	181	317	208.64	386.563	4	3	5
2	256	369	410.333	863.637	4	8	11
3	188	413	275.451	575.931	5	7	4

4	278	483	336.363	692.382	9	9	10
5	123	182	138.154	271.409	4	4	2
6	152	309	194.61	474.747	4	5	6
7	70	70	173.782	259.622	1	2	4
8	130	235	150.439	314.806	6	4	4
9	291	580	378.99	797.838	6	10	6
10	469	926	537.345	1087.02	9	12	8
11	257	407	394.191	958.235	11	8	17
12	50	85	100.042	229.267	1	3	7
13	126	283	261.392	474.943	3	7	5
14	167	329	206.51	643.529	4	4	9
15	580	904	697.612	1291.34	11	16	14
16	255	480	318.907	643.872	6	8	7
17	306	518	353.25	648.63	5	8	9
18	499	821	591.002	981.436	13	17	14
19	168	295	232.551	427.333	6	6	7
20	153	207	194.782	394.978	5	4	6
21	370	622	476.747	999.394	10	12	9


548_hyp_HP_z14c1-2: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 27567)
count
width (px)